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Application of a Novel Method for Machine Performance Degradation Assessment Based on Gaussian Mixture Model and Logistic Regression

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Abstract: The currently prevalent machine performance degradation assessment techniques involve estimating a machine's current condition based upon the recognition of indications of failure features, which entail complete data collected in different conditions. However, failure data are always hard to acquire, thus making those techniques hard to be applied. In this paper, a novel method which does not need failure history data is introduced. Wavelet packet decomposition(WPD) is used to extract features from raw signals, principal component analysis(PCA) is utilized to reduce feature dimensions, and Gaussian mixture model(GMM) is then applied to approximate the feature space distributions. Single-channel confidence value(SCV) is calculated by the overlap between GMM of the monitoring condition and that of the normal condition, which can indicate the performance of single-channel. Furthermore, multi-channel confidence value(MCV), which can be deemed as the overall performance index of multi-channel, is calculated via logistic regression(LR) and that the task of decision-level sensor fusion is also completed. Both SCV and MCV can serve as the basis on which proactive maintenance measures can be taken, thus preventing machine breakdown. The method has been adopted to assess the performance of the turbine of a centrifugal compressor in a factory of Petro-China, and the result shows that it can effectively complete this task. The proposed method has engineering significance for machine performance degradation assessment.

Key words: performance degradation assessment, Gaussian mixture model, logistic regression, proactive maintenance, sensor fusion

1 Introduction

In modern manufacturing processes, well-maintained machines can help improving productivity and cost-effectiveness for manufacturers. However, once certain failure occurs, it may not only disturb the normal working condition of machines, but also result in huge economic losses due to breakdown, so it is of extreme importance to evaluate the health conditions of machines effectively. Usually, machines will undergo a series of measurable degradation processes before failure happens. If we can properly extract the operational features and make health assessment timely and accurately, based on which proactive maintenance measures can be taken, it will be of great benefit to the stable operation of machines^[1].

Currently, considerable techniques have been developed to conduct the task of machine performance degradation assessment^[3]. ZHOU, et al^[4], established a neural network

model for the purpose of postural evaluations of electric equipments. YAN, et al^[5], used logistic regression to assess performance degradation of an elevator system. CAESARENDRA, et al^[6], applied logistic regression and relevance vector machine to the estimation of failure degradation of bearings. POYHONEN, et al^[7], extracted AR model coefficients as features from vibration signals of an induction motor and implemented a fault detection routine with support vector machine. SWANSON^[8] used a fuzzy logic neural network to predict failure of a tensioned steel band with seeded crack growth. ZHU, et al^[9], considered hidden markov model to be applied in the assessment of a cracked rotor dynamic model with decreasing stiffness.

In general, the above models are mainly based on the presumption that data from both normal and failure conditions can be collected. However, historical running data under failure conditions are always difficult to acquire and even unavailable especially for some new products. Therefore, developing the method of assessing performance degradation based only on the normal condition is quite important^[10].

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This paper introduces a method which does not need failure history data to assess machine performance degradation, and the method is applied in the assessment of the real-time performance degradation of the turbine of a centrifugal compressor. The paper is organized as follows: section 2 introduces mathematical theories of the proposed method, section 3 illustrates an engineering case to validate the effectiveness of the method, and finally, this paper reaches conclusions in section 4.

2 Mathematical Theories

2.1 Procedures of the proposed method

The main procedures of the proposed method are described in the following steps.

(1) Feature extraction. Acquire data in both normal and monitoring conditions, extract features with wavelet packet decomposition(WPD) and then conduct feature dimension reduction via principal component analysis(PCA).

(2) SCV calculation. For one single-channel, use Gaussian mixture model(GMM) to approximate the feature distributions of different conditions, and calculate single-channel confidence value(SCV) as a quantitative health indicator of the single-channel performance.

(3) MCV calculation. Complete sensor fusion of multi-channel by calculating multi-channel confidence value(MCV) via logistic regression(LR), which can indicate the comprehensive performance of the turbine.

(4) Decision-making based on SCV or MCV. On the basis of both SCV and MCV, judgment can be made if degradation really happens and if proactive maintenance measures should be taken.

2.2 Feature extraction

2.2.1 WPD

When the performance of a machine degrades, the energy distributions of the vibration signal in different frequency bands will change accordingly compared with those of normal condition, which can be utilized to characterize different operational conditions. Moreover, the vibration signals of machines usually bear the properties of non-linearity and non-stationarity (with frequency component changing over time), while WPD is a powerful tool to capture the dynamic characteristics of the non-stationary signal. So it will be suitable to apply it to extract energies in every frequency band as features^[11], thus acquiring degradation information.

An N -level WPD is conducted as follows.

(1) Suppose (i, j) denotes the j th node in the i th level in the wavelet packet tree, where $i=1, 2, \dots, N, j=0, 1, \dots, 2^{N-1}$.

(2) Reconstruct signals by wavelet packet coefficients and extract signal S_{ij} from each frequency band.

(3) Calculate the total energy of all frequency bands (suppose E_{ij} is the energy of S_{ij}) as follows:

$$E_{ij} = \int |S_{ij}(t)|^2 dt = \sum_{k=1}^n |x_{jk}|^2, \quad (1)$$

where $x_{jk}(j=0, 1, \dots, 7; k=1, 2, \dots, n)$ stands for the amplitude of reconstruction signal S_{ij} .

(4) Construct feature vectors of wavelet packet energy (suppose $N=3$) as follows:

$$\hat{T} = (E_{30}, E_{31}, \dots, E_{37}) / \sum_{j=1}^7 E_{3j}. \quad (2)$$

The output of WPD will be used as the input of PCA module in the following.

2.2.2 PCA

PCA is currently often used to simplify a dataset. If we want to transform a given dataset X of dimension n to a dataset Y of a lower dimension m , we must find the matrix Y that is the Karhunen-Loeve(KL) transform of X in the way of being optimal in a sum-squared error sense^[12], shown as follows:

$$Y = A^T X, \quad (3)$$

where A^T is called KL matrix.

The main steps of the method are briefly shown below:

(1) Calculate the overall mean vector of the dataset X and use it as the origin of new coordinates.

(2) Calculate the autocorrelation matrix of X :

$$R = E(XX^T). \quad (4)$$

(3) Compute the eigenvalues of R : $\lambda_1, \lambda_2, \dots, \lambda_n$ and their corresponding eigenvectors $\varphi_1, \varphi_2, \dots, \varphi_n$, arrange the eigenvalues in the descending order and keep only the first m eigenvectors corresponding to the first m eigenvalues, acquiring $A = (\varphi_1, \varphi_2, \dots, \varphi_m)$.

Dimension reduction is completed by Eq. (3), and therefore feature extraction is completed.

2.3 Single-channel performance assessment

2.3.1 GMM

GMM is the linear combination of a number of Gaussian density functions with different means and covariance matrices. It can theoretically be applied to approximate an arbitrary feature distribution within an arbitrary accuracy^[13]. The mathematic model of GMM is described as below:

$$H(x) = \sum_{i=1}^I p_i h(x; \theta_i), \quad (5)$$

$$h(x; \theta_i) = \frac{1}{(2\pi)^{d/2} |\sigma_i|^{1/2}} \exp\left[-\frac{1}{2}(x - \mu_i)' \sigma_i^{-1} (x - \mu_i)\right],$$

where $h(\mathbf{x}; \boldsymbol{\theta}_i)$ is a single Gaussian function, $\mathbf{x}$ is the d -dimensional feature vector, I is number of mixtures, and $\mathbf{p}_i$ is the prior vector, satisfying $\sum_{i=1}^I \mathbf{p}_i = 1$. $\boldsymbol{\theta}_i$ stands for the

parameters of the i th Gaussian function, including mean vector $\boldsymbol{\mu}_i$ and covariance matrix $\boldsymbol{\sigma}_i$.

The unknown parameters could be estimated using the expectation maximization(EM) algorithm, the main steps of which is shown as follows.

(1) (E step) Using Bayes' theorem, calculate the posterior probability of model i , given a data vector $\mathbf{x}_n$ and current model parameter $\boldsymbol{\theta}$

$$p(i | \mathbf{x}_n, \boldsymbol{\theta}) = \frac{\mathbf{p}_i p(\mathbf{x}_n | \boldsymbol{\theta}_i)}{\sum_{j=1}^I \mathbf{p}_j p(\mathbf{x}_n | \boldsymbol{\theta}_j)}. \quad (6)$$

(2) (M step) Maximum likelihood(ML) re-estimation of model parameters is given by

$$\hat{\mathbf{p}}_i = \frac{1}{N} \sum_{n=1}^N p(i | \mathbf{x}_n, \boldsymbol{\theta}), \quad (7)$$

$$\hat{\boldsymbol{\mu}}_i = \frac{\sum_{n=1}^N p(i | \mathbf{x}_n, \boldsymbol{\theta}) \mathbf{x}_n}{\sum_{n=1}^N p(i | \mathbf{x}_n, \boldsymbol{\theta})}, \quad (8)$$

$$\hat{\boldsymbol{\sigma}}_i = \frac{\sum_{n=1}^N p(i | \mathbf{x}_n, \boldsymbol{\theta}) (\mathbf{x}_n - \hat{\boldsymbol{\mu}}_i)(\mathbf{x}_n - \hat{\boldsymbol{\mu}}_i)'}{\sum_{n=1}^N p(i | \mathbf{x}_n, \boldsymbol{\theta})}. \quad (9)$$

By repeating E step and M step in an iterative procedure, the calculation converges to a stable solution that represents the maximum likelihood solution of the problem, thus getting converged mean, covariance matrix and prior vectors.

2.3.2 Bayesian Information Criterion

When using GMM to approximate the feature distribution, how to properly choose the number of the mixture is critical, or the problems of underfitting or overfitting will arise. Bayesian Information Criterion (BIC) is introduced to address this issue^[14].

BIC defines a log-likelihood function and penalty term, and its mathematical description given GMM is depicted as follows:

$$BIC(H_j | D) = -2 \times \ln(P(D | \hat{\boldsymbol{\theta}}_j, H_j)) + k \times \ln(n), \quad (10)$$

where H_j is the j th candidate model, D is the training feature, $\ln(P(D | \hat{\boldsymbol{\theta}}_j, H_j))$ is the maximized log-likelihood of the j th candidate model, k is the number of the

parameters to be estimated and n is the size of the feature. The candidate GMM with the lowest BIC score can be deemed as the best model.

2.3.3 Calculating SCV

For a single channel, after establishing the GMMs for normal condition and monitoring condition separately, SCV can be computed by the overlap of the GMMs according to Eq. (11) to quantitatively indicate the health performance of one channel^[15].

$$V_{sc} = \frac{\int g_1(\mathbf{x}) g_2(\mathbf{x}) d\mathbf{x}}{\sqrt{\int (g_1(\mathbf{x}))^2 d\mathbf{x}} \sqrt{\int (g_2(\mathbf{x}))^2 d\mathbf{x}}}. \quad (11)$$

Where $g_1(\mathbf{x})$ and $g_2(\mathbf{x})$ are GMMs which represent the normal and monitoring condition, respectively, and the SCV is between 0 and 1. It is assumed that the higher the SCV is, the higher the overlap between the two conditions will be, and therefore the performance of the monitoring condition will be close to normal. On the contrary, if SCV becomes lower than a predefined value, it means some degradation may have occurred, and the lower it is, the more degraded it may be, thus requiring timely maintenance measures. To make it more clear, the illustration of SCV in a 2-dimensional feature space is shown in Fig. 1.

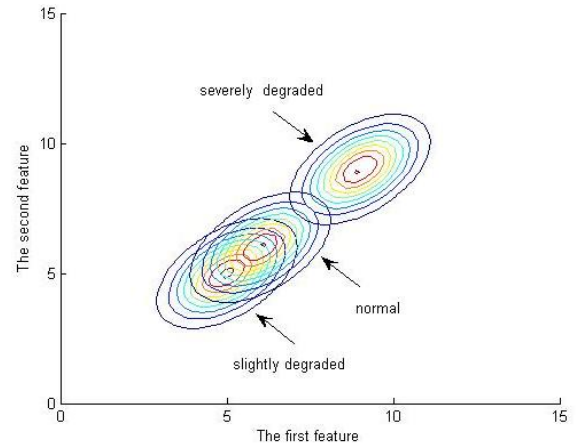


Fig. 1. Illustration of SCV

Based on the assumptions above, we can set a performance health threshold according to practical application experience (such as 0.8). If SCV is under this threshold, we can judge that some abnormality has occurred in this channel.

2.4 Multi-channel performance assessment

After SCV from each channel is calculated, the problem on how to judge the overall performance of the machines with multi-channel arises, which makes the sensor fusion a must. There are several techniques to achieve this goal and one of them is decision-level sensor fusion, which is defined by the Joint Directors of Laboratories(JDL)

standard for sensor fusion^[16]. In this section, LR is investigated to accomplish this task.

2.4.1 LR

LR is part of a category of statistical models called generalized linear models. LR allows one to predict a discrete outcome from a set of variables that may be continuous, discrete, dichotomous, or a mix of any of these. The logistic function defines a probability:

$$P(y=1|\mathbf{x}) = \frac{1}{1 + \exp(-(\alpha + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_k x_k))}, \quad (12)$$

where $\mathbf{x} = (x_1, x_2, \dots, x_k)^T$ is the input with k dimension, and P is the output between 0 and 1.

To train the model, a group of sample inputs $\{\mathbf{x}\}$ are observed, and the corresponding probabilities $\{P(\mathbf{x})\}$ are estimated from statistics or specified subjectively according to needs, so that the parameters $\alpha, \beta_1, \dots, \beta_k$ can be obtained by using maximum likelihood estimation(MLE) method, which finds the parameter values that maximize the probability of the observed data by iterative steps.

2.4.2 Calculating MCV

Theoretically, both normal and faulty data are needed to train the LR model. However, due to the lack of historical data and the need to fulfill the specific tasks, we can specify the input and output of the model based on the expertise and experience of the practitioners.

After the parameters of LR are obtained by MLE, SCVs of each channel will be used as inputs of Eq. (12), and the output P is defined as MCV, which can signify the comprehensive performance of the machine.

3 An Engineering Application Case

3.1 System Setup

Centrifugal compressors are widely used and play key roles in the manufacturing processes of various types of equipments. However, there may occur some known or unknown faults that will harm their stable operation, thus causing huge economic losses, so that it will be of extreme importance to conduct real time performance degradation assessment and determine if certain abnormality has happened, thus preventing degradation from deteriorating. In this study, the data acquired from a centrifugal compressor will be used to validate the method presented in this study.

Main structure of 102J type centrifugal compressor is shown in Fig. 2, which mainly comprises a turbine, a high-pressure cylinder and a low-pressure cylinder. Specifically, to monitor the real condition of the turbine, data is acquired by accelerometers installed both horizontally in channel 1H, channel 2H and vertically in channel 1V and channel 2V. Rotation speed of the rotor is approximately 9 106 r/min, and data sampling frequency is

5 120 Hz, with each sampling period acquiring 1 024 data points.

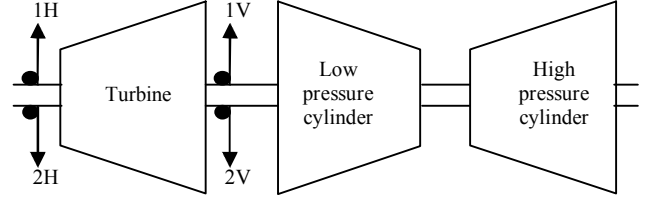


Fig. 2. Main structure of 102J type centrifugal compressor

As shown in Fig. 3, raw signals acquired from three different time intervals in channel 1H are presented, which are baseline condition (acquired from stable operational condition), normal condition (also acquired from stable operational condition) and degraded condition (acquired from degraded operational condition due to wide gaps between bearings in the turbine) separately, each interval comprising $1\,024 \times 60$ data points.

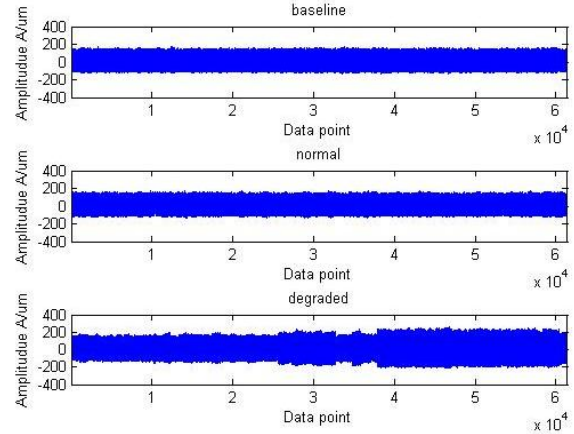


Fig. 3. Raw signals of 3 different conditions in channel 1H

Simply from the time domain waveform, it is not very easy to judge if degradation really happens, let alone quantitatively assessing its extent of degradation. As a result, we try to resolve this problem in the following.

3.2 Performance assessment results

Firstly, 3-level WPD is used to extract the 8-dimensional feature from raw signals in one sampling period (which contains 1 024 data points), so we can get an 8×60 feature matrix characterizing each condition in Fig. 3.

Secondly, PCA is employed to reduce the dimension of features from 8 to 3, ensuring that the 3 component features containing more than 90% of the variation information, and therefore the feature spaces of the three conditions can be drawn in Fig. 4 and Fig. 5.

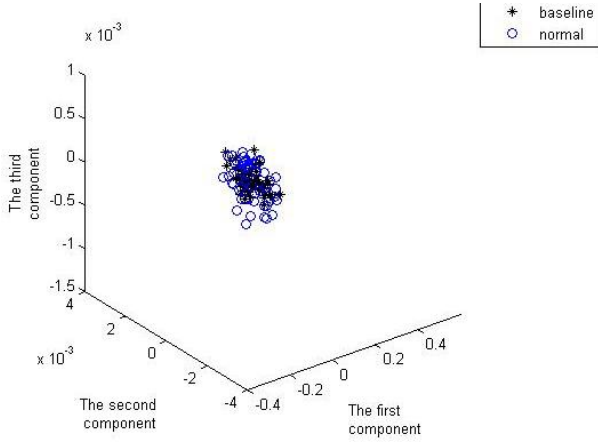


Fig. 4. Baseline feature space vs. normal feature space

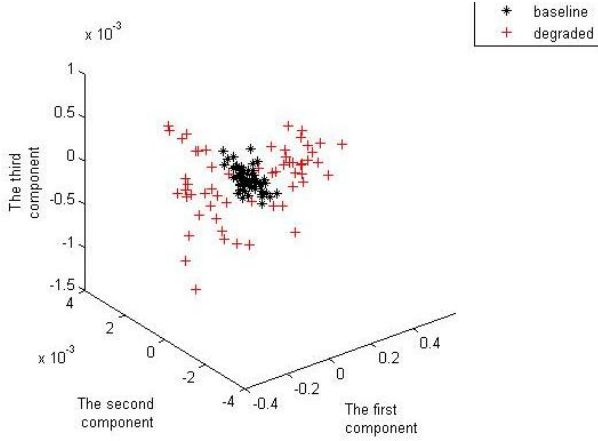


Fig. 5. Baseline feature space vs. degraded feature space

Thirdly, GMM is applied to approximate the density distributions of three feature spaces and the parameters of each GMM can be obtained through EM algorithm. In addition, the number of mixture of each GMM is determined by BIC. Just as shown in Fig. 6, the BIC scores for different candidate GMMs (with number of mixture from 1 to 6) are computed by Eq. (10), and the optimal number of mixture could be 1, 1, and 3 for each GMM in each condition.

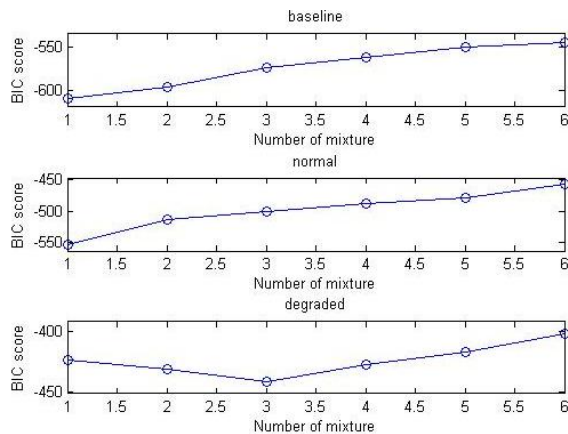


Fig. 6. BIC scores for each GMM of the 3 conditions

As shown in Fig. 4, the feature space distribution of

normal condition and that of the baseline condition extensively overlap, and by using Eq. (11), the calculated SCV is 0.999 7, which means the performance is normal. However, as shown in Fig. 5, the feature space distribution of degradation condition deviates widely from that of the baseline condition, and the calculated SCV by using Eq. (11) is 0.761 4, which means the performance is degraded. Consequently, it has been confirmed that the computation result matches the reality that there actually exist wide gaps between bearings in the turbine. Moreover, 0.80 can be set as the single-channel health threshold.

Similarly, SCVs of channel 1V, channel 2H, and channel 2V can also be calculated, and the results are shown in Table 1.

Table 1. SCV of each channel in different conditions

Channel	Normal condition	Degraded condition
1H	0.999 7	0.761 4
1V	0.997 9	0.806 7
2H	0.996 2	0.700 1
2V	0.999 4	0.730 7

Next, in order to assess the overall performance of the turbine, LR is utilized to conduct decision-level sensor fusion of the four channels. Given calculation results and facts above, we define the training input and training output of LR as Table 2 shows, so that the parameter of the model can be acquired through MLE algorithm, and the model is expressed as

$$V_{MC} = 1 / [1 + \exp(-(2.402 1 + 1.230 3x_1 + 1.230 3x_2 + 1.230 3x_3 + 1.230 3x_4))]. \quad (13)$$

Table 2. Input and output data to train LR

Condition	1H	1V	2H	2V	MCV
Normal	0.95	0.95	0.95	0.95	0.95
Slightly degraded	0.80	0.80	0.80	0.80	0.80
Severely degraded	0.50	0.50	0.50	0.50	0.50
Fault	0.30	0.30	0.30	0.30	0.30

Finally, MCV can be calculated by using SCVs from Table 1 as inputs, and the results are 0.924 9 and 0.783 7 for normal and degraded condition respectively. We can find that MCVs in both conditions are in accordance with the fact. In normal condition, it is much higher than 0.80, while in degraded condition it falls lower than 0.80, which can demonstrate the overall performance of the turbine. Moreover, the overall health threshold can be set as 0.80, under which the degradation can be thought to have happened. Consequently, the method introduced in this paper has been validated and proactive maintenance can be taken to avoid machine breakdown.

4 Conclusions

(1) An engineering application case has validated a novel method for machine performance degradation assessment,

which does not need failure history data.

(2) SCV and MCV serve as the quantitative indicator of the current performance of the turbine, either partially or comprehensively. Once the value become lower than the health threshold, we can infer that some degradation or incipient failure has occurred, based on which early warning strategy can be activated and proactive maintenance measures should be taken.

(3) The proposed method is generic and can be used in the performance degradation assessment of other machines.

In future, some improvements can be made from various angles as follows. Feature extraction and dimension reduction can be improved by other more robust algorithms; The approximation of feature space via GMM adopts traditional EM algorithm and BIC, while in some cases it may reach a local convergence. To avoid this, other improvements should be made; If SCV is lower than the original health threshold while the machine is still in stable condition, data to train the baseline GMM should be updated and the threshold should be adjusted; As to sensor fusion, LR is a tentative exploration in this paper, and many other methods can be utilized.

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